

KCET – Mathematics – 2018
VERSION CODE: B

Question 1: The distance between the foci of a hyperbola is 16 and its eccentricity is $\sqrt{2}$. Its equation is

- (A) $x^2 - y^2 = 32$
- (B) $\frac{x^2}{4} - \frac{y^2}{9} = 1$
- (C) $2x^2 - 3y^2 = 7$
- (D) $y^2 - x^2 = 32$

Answer 1: (A)

Question 2: The number of ways in which 5 girls and 3 boys can be seated in a row so that no two boys are together is

- (A) 14040
- (B) 14440
- (C) 14000
- (D) 14400

Answer 2: (D)

Question 3: If a, b, c and three consecutive terms of an AP and x, y, z are three consecutive terms of GP, then the value of $x^{b-c}, y^{c-a}, z^{a-b}$ is

- (A) 0
- (B) xyz
- (C) -1
- (D) 1

Answer 3: (D)

Question 4: The value of $\lim_{x \rightarrow 0} \frac{|x|}{x}$ is

- (A) 1
- (B) -1
- (C) 0
- (D) Does not exist

Answer 4: (D)

Question 5: Let $f(x) = x - \frac{1}{x}$ then $f'(-1)$ is

- (A) 0
- (B) 2
- (C) 1
- (D) -2

Answer 5: (B)

Question 6: The negation of the statement “72 is divisible by 2 and 3” is

- (A) 72 is not divisible by 2 or 72 is not divisible by 3
- (B) 72 is divisible by 2 or 72 is divisible by 3
- (C) 72 is divisible by 2 and 72 is divisible by 3
- (D) 72 is not divisible by 2 and 3

Answer 6: (A)

Question 7: The probability of happening of an event A is 0.5 and that of B is 0.3. If A and B are mutually exclusive events, then the probability of neither A nor B is

- (A) 0.4
- (B) 0.5
- (C) 0.2
- (D) 0.9

Answer 7: (C)

Question 8: In a simultaneous throw of a pair of dice, the probability of getting a total more than 7 is

- (A) $\frac{7}{12}$
- (B) $\frac{5}{36}$
- (C) $\frac{5}{12}$
- (D) $\frac{7}{36}$

Answer 8: (C)

Question 9: If A and B are mutually exclusive events, given that $P(A) = \frac{3}{5}$, $P(B) = \frac{1}{5}$, then, P(A

or B) is

- (A) 0.8
- (B) 0.6
- (C) 0.4
- (D) 0.2

Answer 9: (A)

Question 10: Let $f, g : R \rightarrow R$ be two function defined as $f(x) = |x| + x$ and $g(x) = |x| - x \forall x \in R$.
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(A) 0

(B) $4x$

(C) $-4x$

(D) $2x$

Answer 10: (C)

Question 11: A is set having 6 distinct elements. The number of distinct functions from A to A which are not bisections is

(A) $6! - 6$

(B) $6^6 - 6$

(C) $6^6 - 6!$

(D) $6!$

Answer 11: (C)

Question 12: Let $f : R \rightarrow R$ defined by $f(x) = \begin{cases} 2x & : x > 3 \\ x^2 & : 1 < x \leq 3 \\ 3x & : x \leq 1 \end{cases}$. Then $f(-1) + f(2) + f(4)$

(A) 9

(B) 14

(C) 5

(D) 10

Answer 12: (A)

Question 13: If $\sin^{-1} x + \cos^{-1} y = \frac{2\pi}{5}$, then $\cos^{-1} x + \sin^{-1} y$ is

(A) $\frac{2\pi}{5}$

(B) $\frac{3\pi}{5}$

(C) $\frac{4\pi}{5}$

(D) $\frac{3\pi}{10}$

Answer 13: (B)

Question 14: The value of the expression $\tan\left(\frac{1}{2}\cos^{-1}\frac{2}{\sqrt{5}}\right)$ is

(A) $2 - \sqrt{5}$

(B) $\sqrt{5} - 2$

(C) $\frac{\sqrt{5} - 2}{2}$

(D) $5 - \sqrt{2}$

Answer 14: (B)

Question 15: If $A = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$, then $A^n = 2^k A$, where $k =$

(A) 2^{n-1}

(B) $n + 1$

(C) $n - 1$

(D) $2(n - 1)$

Answer 15: (D)

Question 16: If $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, then the values of x and y respectively are,

(A) $-3, -1$

(B) $1, 3$

(C) $3, 1$

(D) $-1, 3$

Answer 16: (D)

Question 17: If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then $AA' =$

(A) A

(B) Zero matrix

(C) A'

(D) I

Answer 17: (D)

Question 18: If $x, y, z \in \mathbb{R}$, then the value of determinant $\begin{vmatrix} (5^x + 5^{-x})^2 & (5^x - 5^{-x})^2 & 1 \\ (6^x + 6^{-x})^2 & (6^x - 6^{-x})^2 & 1 \\ (7^x + 7^{-x})^2 & (7^x - 7^{-x})^2 & 1 \end{vmatrix}$ is

- (A) 10
- (B) 12
- (C) 1
- (D) 0

Answer 18: (D)

Question 19: The value of determinant $\begin{vmatrix} a-b & b+c & a \\ b-a & c+a & b \\ c-a & a+b & c \end{vmatrix}$ is

- (A) $a^3 + b^3 + c^3$
- (B) $3abc$
- (C) $a^3 + b^3 + c^3 - 3abc$
- (D) $a^3 + b^3 + c^3 + 3abc$

Answer 19: (B)

Question 20: If (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are the vertices of a triangle whose area is 'k' square units, then

$\begin{vmatrix} x_1 & y_1 & 4 \\ x_2 & y_2 & 4 \\ x_3 & y_3 & 4 \end{vmatrix}^2$ is

- (A) $32k^2$
- (B) $16k^2$
- (C) $64k^2$
- (D) $48k^2$

Answer 20: (C)

Question 21: Let A be a square matrix of order 3×3 , the $|5A| =$

- (A) $5|A|$
- (B) $125|A|$
- (C) $25|A|$
- (D) $15|A|$

Answer 21: (B)

Question 22: If $f(x) = \begin{cases} \frac{\sqrt{1+kx} - \sqrt{1-kx}}{x} & \text{if } -1 \leq x < 0 \\ \frac{2x+1}{x-1} & \text{if } 0 \leq x \leq 1 \end{cases}$ is continuous at $x = 0$, then the value of

k

(A) $k = 1$

(B) $k = 1$

(C) $k = 0$

(D) $k = 2$

Answer 22: (B)

Question 23: If $\cos y = x \cos(a + y)$ with $\cos \neq \pm 1$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{\sin a}{\cos^2(a + y)}$

(B) $\frac{\cos^2(a + y)}{\sin a}$

(C) $\frac{\cos a}{\sin^2(a + y)}$

(D) $\frac{\cos^2(a + y)}{\cos a}$

Answer 23: (B)

Question 24: If $f(x) = |\cos x - \sin x|$, then $f'\left(\frac{\pi}{6}\right)$ is equal to

(A) $-\frac{1}{2}(1 + \sqrt{3})$

(B) $\frac{1}{2}(1 + \sqrt{3})$

(C) $-\frac{1}{2}(1 - \sqrt{3})$

(D) $\frac{1}{2}(1 - \sqrt{3})$

Answer 24: (A)

Question 25: If $\sqrt{x + \sqrt{x + \sqrt{x + \dots + \infty}}}$, then $\frac{dy}{dx} =$

(A) $\frac{1}{y^2 - 1}$

(B) $\frac{1}{2y + 1}$

(C) $\frac{2y}{y^2 - 1}$

(D) $\frac{1}{2y - 1}$

Answer 25: (D)

Question 26: If $f(x) = \begin{cases} \frac{\log_e x}{x-1}; & x \neq 1 \\ k; & x = 1 \end{cases}$ is continuous at $x = 1$, then the value of k is

(A) e

(B) 1

(C) -1

(D) 0

Answer 26: (B)

Question 27: Approximate change in the volume V of a cube of side x metres caused by increasing the side by 3% is

(A) $0.09 x^3 \text{m}^3$

(B) $0.03 x^3 \text{m}^3$

(C) $0.06 x^3 \text{m}^3$

(D) $0.04 x^3 \text{m}^3$

Answer 27: (A)

Question 28: The maximum value of $\left(\frac{1}{x}\right)^x$ is

(A) e

(B) e^e

(C) $e^{1/e}$

(D) $\left(\frac{1}{e}\right)^{1/e}$

Answer 28: (C)

Question 29: $f(x) = x^x$ has stationary point at

(A) $x = e$

(B) $x = \frac{1}{e}$

(C) $x = 1$

(D) $x = \sqrt{e}$

Answer 29: (B)

Question 30: The maximum area of rectangle inscribed in the circle $(x+1)^2 + (y-3)^2 = 64$

(A) 64 sq. units

(B) 72 sq. units

(C) 128 sq. units

(D) 8 sq. units

Answer 30: (C)

Question 31: $\int \frac{1}{1+e^x} dx$ is equal to

(A) $\log_e \left(\frac{e^x+1}{e^x} \right) + e$

(B) $\log_e \left(\frac{e^x-1}{e^x} \right) + e$

(C) $\log_e \left(\frac{e^x-1}{e^x} \right) + e$

(D) $\log_e \left(\frac{e^x-1}{e^x} \right) + e$

Answer 31: (C)

Question 32: $\int \frac{1}{\sqrt{3-6x-9x^2}} dx$ is equal to

(A) $\sin^{-1} \left(\frac{3x+1}{2} \right) + c$

(B) $\sin^{-1} \left(\frac{3x+1}{6} \right) + c$

(C) $\frac{1}{3} \sin^{-1} \left(\frac{3x+1}{2} \right) + c$

(D) $\sin^{-1} \left(\frac{2x+1}{3} \right) + c$

Answer 32: (C)

Question 33: $\int e^{\sin x} \cdot \left(\frac{\sin x + 1}{\sec x} \right) dx$ is equal to

- (A) $\sin x \cdot e^{\sin x} + c$
- (B) $\cos x \cdot e^{\sin x} + c$
- (C) $e^{\sin x} + c$
- (D) $e^{\sin x} (\sin x + 1) + c$

Answer 33: (A)

Question 34: $\int_{-2}^3 |x \cos \pi x| dx$ is equal to

- (A) $\frac{8}{\pi}$
- (B) $\frac{4}{\pi}$
- (C) $\frac{2}{\pi}$
- (D) $\frac{1}{\pi}$

Answer 34: (A)

Question 35: $\int_0^1 \frac{dx}{e^x + e^{-x}}$ is equal to

- (A) $\frac{\pi}{4} \tan^{-1}(e)$
- (B) $\tan^{-1}(e) - \frac{\pi}{4}$
- (C) $\tan^{-1}(e) + \frac{\pi}{4}$
- (D) $\tan^{-1}(e)$

Answer 35: (B)

Question 36: $\int_0^{1/2} \frac{dx}{(1+x^2)\sqrt{1-x^2}}$ is equal to

- (A) $\frac{1}{2} \tan^{-1} \sqrt{\frac{2}{3}}$
- (B) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3}{\sqrt{2}} \right)$
- (C) $\frac{\sqrt{2}}{2} \tan^{-1} \left(\frac{3}{2} \right)$
- (D) $\frac{\sqrt{2}}{2} \tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$

Answer 36: (A)

Question 37: The area of region bounded by the curve $y = \cos x$ between $x = 0$ and $x = \pi$ is

- (A) 1 sq. unit
- (B) 4 sq. units
- (C) 2 sq. units
- (D) 3 sq. units

Answer 37: (C)

Question 38: The area bounded by the line $y = x$, x -axis and ordinates $x = -1$ and $x = 2$ is

- (A) $\frac{3}{2}$
- (B) $\frac{5}{2}$
- (C) 2
- (D) 3

Answer 38: (B)

Question 39: The degree and the order of the differential equation $\frac{d^2 y}{dx^2} = \sqrt[3]{1 + \left(\frac{dy}{dx} \right)^2}$

respectively are

- (A) 2 and 3
- (B) 3 and 2
- (C) 2 and 2
- (D) 3 and 3

Answer 39: (B)

Question 40: The solution of the differential equation $x \frac{dy}{dx} - y = 3$ represents a family of

- (A) Straight lines
- (B) Circles
- (C) Parabolas
- (D) Ellipses

Answer 40: (A)

Question 41: The integrating factor of $\frac{dy}{dx} + y = \frac{1+y}{x}$ is

- (A) xe^x
- (B) $xe^{1/x}$
- (C) $\frac{e^x}{x}$
- (D) $\frac{x}{e^x}$

Answer 41: (C)

Question 42: If $|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = 144$ and $|\vec{a}| = 4$ then, $|\vec{b}|$ is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Answer 42: (C)

Question 43: If $\vec{a}$ and $\vec{b}$ are mutually perpendicular unit vectors, then $(3\vec{a} + 2\vec{b}) \cdot (5\vec{a} - 6\vec{b}) =$

- (A) 5
- (B) 3
- (C) 6
- (D) 12

Answer 43: (B)

Question 44: If the vectors $a\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ are coplanar ($a \neq b \neq c \neq 1$), then the value of $abc - (a + b + c) =$

- (A) 2
- (B) -2
- (C) 0
- (D) -1

Answer 44 : (B)

Question 45: If $\vec{a} = \hat{i} + \lambda\hat{j} + 2\hat{k}; b = \mu\hat{i} + \hat{j} - \hat{k}$ are orthogonal and $|\vec{a}| = |\vec{b}|$ the $(\lambda, \mu) =$

- (A) $\left(\frac{1}{4}, \frac{7}{4}\right)$
- (B) $\left(\frac{7}{4}, \frac{1}{4}\right)$
- (C) $\left(\frac{1}{4}, \frac{9}{4}\right)$
- (D) $\left(\frac{-1}{4}, \frac{9}{4}\right)$

Answer 45: (A)

Question 46: The image of the point (1, 6, 3) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is

- (A) (1, 0, 7)
- (B) (7, 0, 1)
- (C) (2, 7, 0)
- (D) (-1, -6, -3)

Answer 46: (A)

Question 47: The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is

- (A) 0°
- (B) 45°
- (C) 90°
- (D) 30°

Answer 47: (C)

Question 48: The value of k such that the line $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies on the plane $2x - 4y - z =$
7 is

- (A) -7
- (B) 4
- (C) -4
- (D) 7

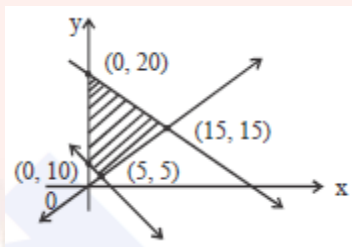
Answer 48: (D)

Question 49: The locus represented by $xy + yz = 0$ is

- (A) A pair of perpendicular lines
- (B) A pair of parallel lines
- (C) A pair of parallel planes
- (D) A pair of perpendicular planes

Answer 49: (D)

Question 50: The feasible region of an LPP is shown in the figure. If $z = 3x + 9y$, then the minimum value of z occurs at



- (A) (5, 5)
- (B) (0, 10)
- (C) (0, 20)
- (D) (15, 15)

Answer 50: (A)

Question 51: For the LPP; maximize $z = x + 4y$ subject to the constraint $x + 2y \leq 2, x + 2y \geq 8, x, y \geq 0$

- (A) $Z_{\max} = 4$
- (B) $Z_{\max} = 8$
- (C) $Z_{\max} = 16$
- (D) Has no feasible solution

Answer 51: (D)

Question 52: For the probability distribution given by

$X=x_1$	0	1	2
P_1	$\frac{25}{36}$	$\frac{5}{18}$	$\frac{1}{36}$

the standard deviation (σ) is

- (A) $\sqrt{\frac{1}{3}}$

(B) $\frac{1}{3}\sqrt{\frac{5}{2}}$

(C) $\sqrt{\frac{5}{36}}$

(D) None of these

Answer 52: (B)

Question 53: A bag contains 17 tickets numbered from 1 to 17. A ticket is drawn at random, then another ticket is drawn without replacing the first one. The probability that both the tickets may show even numbers is

(A) $\frac{7}{34}$

(B) $\frac{8}{17}$

(C) $\frac{7}{16}$

(D) $\frac{7}{17}$

Answer 53: (A)

Question 54: A flashlight has 10 batteries out of which 4 are dead. If 3 batteries are selected without replacement and tested, then the probability that all 3 are dead is

(A) $\frac{1}{30}$

(B) $\frac{2}{8}$

(C) $\frac{1}{15}$

(D) $\frac{1}{10}$

Answer 54: (A)

Question 55: If $|x+5| \geq 10$ then,

(A) $x \in (-15, 5]$

(B) $x \in (-5, 5)$

(C) $x \in (-\infty, -15] \cup [5, \infty)$

(D) $x \in [-\infty, -15] \cup [5, \infty)$

Answer 55: (C)

Question 56: Everybody in a room shakes hands with everybody else. The total number of handshakes is 45. The total number of persons in the room is

- (A) 9
- (B) 10
- (C) 5
- (D) 15

Answer 56: (B)

Question 57: The constant term in the expansion of $\left(x^2 - \frac{1}{x^2}\right)^{16}$ is

- (A) ${}^{16}C_8$
- (B) ${}^{16}C_7$
- (C) ${}^{16}C_9$
- (D) ${}^{16}C_{10}$

Answer 57: (A)

Question 58: If $P(n)$: " $2^{2n} - 1$ is divisible by k for all $n \in \mathbb{N}$ " is true, then the value of ' k ' is

- (A) 6
- (B) 3
- (C) 7
- (D) 2

Answer 58: (B)

Question 59: The equation of the line parallel to the line $3x - 4y + 2 = 0$ and passing through $(-2, 3)$ is

- (A) $3x - 4y + 18 = 0$
- (B) $3x - 4y - 18 = 0$
- (C) $3x + 4y + 18 = 0$
- (D) $3x + 4y - 18 = 0$

Answer 59: (A)

Question 60: If $\left(\frac{1-i}{1+i}\right)^{96} = a + ib$ then (a, b) is

- (A) (1, 1)
- (B) (1, 0)
- (C) (0, 1)
- (D) (0, -1)

Answer 60: (B)